



Preparation of Salts

Contents

- * Preparing Soluble Salts
- * Preparing Insoluble Salts
- * Solubility Rules
- * Hydrated & Anhydrous Salts



Preparing soluble salts

How to name a salt

- The name of salt has two parts:
 1. The first part comes from the **metal**, **metal** oxide or **metal** carbonate used in the reaction
 2. The second part comes from the **acid**
- The name of the salt can be determined by looking at the reactants
 - For example, hydrochloric acid always produces salts that end in chloride and contain the **chloride** ion, Cl^-
- Other examples:
 - **Sodium** hydroxide reacts with hydro**chloric** acid to produce sodium chloride
 - **Zinc** oxide reacts with **sulfuric** acid to produce **zinc sulfate**

What is a salt?

- A salt is a compound that is formed when the **hydrogen atom** in an acid is replaced by a **metal** or **ammonium ion**
 - For example, replacing H in HCl with potassium gives potassium chloride, KCl
- Salts have many uses including:
 - Fertilisers
 - Batteries
 - Cleaning products
 - Healthcare products
 - Fungicides
- The method used depends on:
 - The **solubility** of the salt being prepared
 - Whether the base is **insoluble** or **soluble (alkali)**

Preparing soluble salts

- There are two main methods of preparing a soluble salt:
- **Method A: Using excess solid reactant**
 - This includes three reactions:



1. Using excess metal
2. Using excess insoluble base
3. Using excess insoluble carbonate

■ Method B: Using an alkali (by titration)

- This is for soluble bases such as sodium hydroxide
- The method used depends on:
 - Whether the base is **insoluble** or **soluble (alkali)**
 - Whether a **metal** is used

Method A: Using excess solid reactant

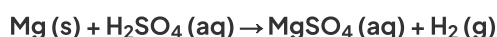
This method is used when the reactant is a **solid in excess**. Any unreacted solid is removed by filtration, then the salt is crystallised.

Method A1: Using excess metal

- This is for metals that react with dilute acids
1. Add dilute acid to a beaker
 2. Add the metal in small pieces while stirring until no more reacts
 - This means that the metal is in excess
 3. Filter the mixture to remove the excess metal
 4. Transfer the filtrate (salt solution) to an evaporating basin
 5. Heat the evaporating basin gently until the solution is concentrated
 - Check the solution is saturated by dipping a cold, glass rod into the solution and seeing if crystals form on the end
 6. Leave the basin in a warm place to crystallize
 7. If necessary, decant any excess liquid
 8. Dry the crystals with filter paper

Example reaction - preparing magnesium sulfate crystals:

magnesium + sulfuric acid → magnesium sulfate + hydrogen



Method A2: Using excess insoluble base

- This is for insoluble bases such as metal oxides
1. Warm dilute acid gently in a beaker
 2. Add the insoluble base slowly while stirring until no more reacts



Your notes

- This means that the insoluble base is in excess
3. Filter the mixture to remove the excess insoluble base
 4. Transfer the filtrate (salt solution) to an evaporating basin
 5. Heat the evaporating basin gently until the solution is concentrated
 - Check the solution is saturated by dipping a cold, glass rod into the solution and seeing if crystals form on the end
 6. Leave the basin in a warm place to crystallize
 7. If necessary, decant any excess liquid
 8. Dry the crystals with filter paper

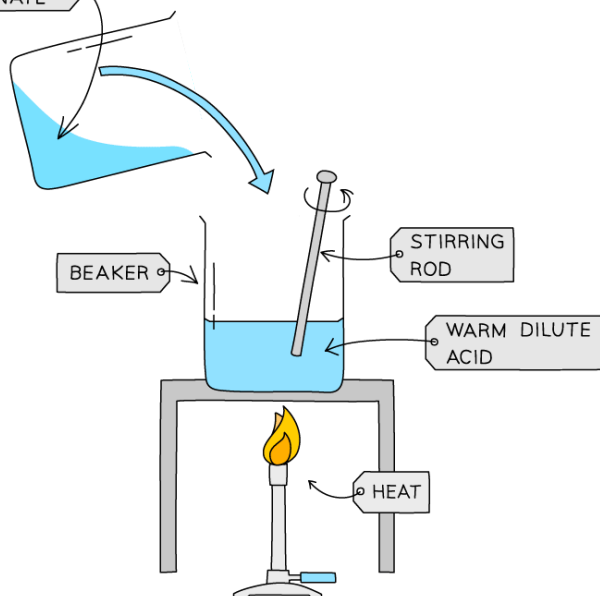


Your notes

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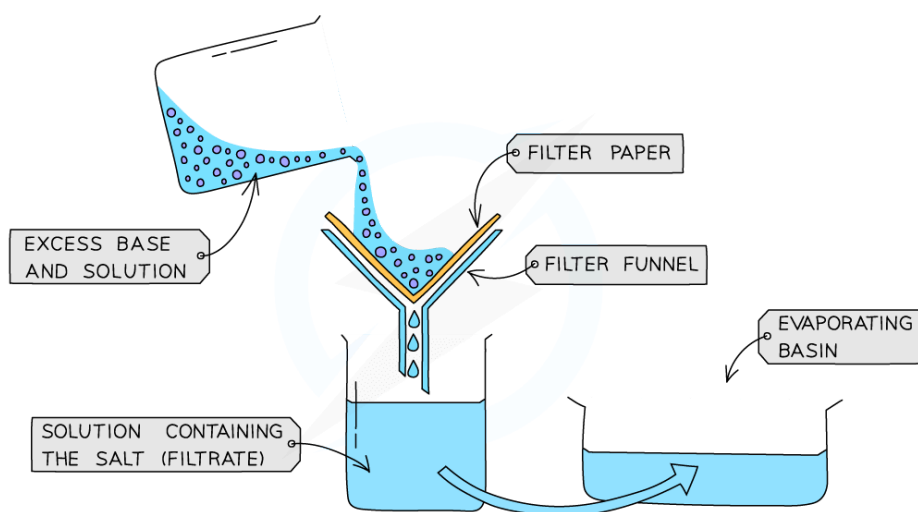
HEAT ACID UNTIL WARM, THEN ADD METAL / BASE / CARBONATE, STIRRING CONSTANTLY UNTIL IT STOPS DISAPPEARING

INSOLUBLE METAL /
BASE / CARBONATE



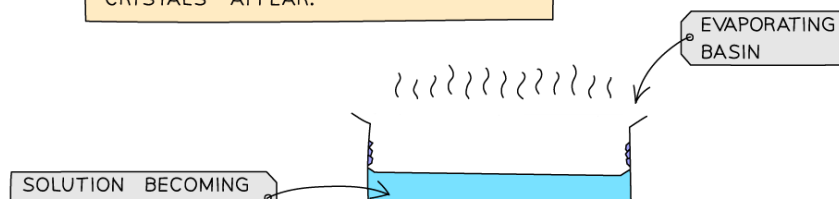
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FILTER MIXTURE TO REMOVE EXCESS BASE, TRANSFER SOLUTION TO EVAPORATING BASIN.

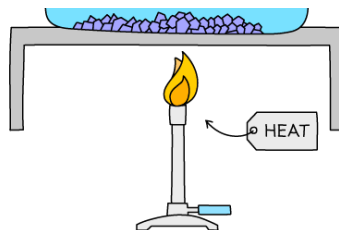


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EVAPORATING WATER FROM SOLUTION USING A BUNSEN BURNER UNTIL CRYSTALS APPEAR.

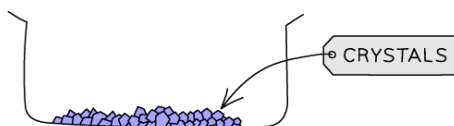


MORE CONCENTRATED



4

REMOVE EVAPORATING BASIN FROM HEAT AND ALLOW FILTRATE TO DRY AND CRYSTALLISE



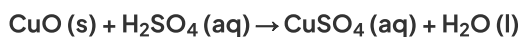
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Diagram showing the preparation of soluble salts

- Example reaction - preparing copper(II) sulfate:

copper(II) oxide + sulfuric acid → copper(II) sulfate + water



Method A3: Using excess insoluble carbonate

- This is for metal carbonates that do not dissolve in water
- The method is exactly the same as using excess insoluble base:
 - Warm dilute acid gently in a beaker
 - Add the insoluble carbonate slowly while stirring until no more reacts (carbonate in excess)
 - Filter to remove the excess insoluble carbonate
 - Transfer the filtrate (salt solution) to an evaporating basin
 - Heat the evaporating basin gently until the solution is concentrated
 - Leave to crystallize, decant any excess liquid, and dry the crystals
- Example reaction - preparing calcium chloride:

calcium carbonate + hydrochloric acid → calcium chloride + carbon dioxide + water



Method B: Using an alkali (by titration)

- This is for soluble bases such as sodium hydroxide



Your notes



Your notes

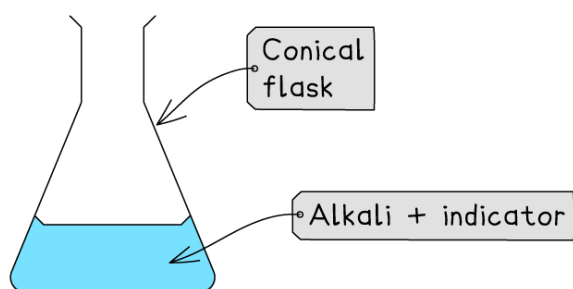
1. Use a pipette to place alkali in a conical flask
2. Add a few drops of indicator (e.g. phenolphthalein)
3. Fill a burette with the acid and record the starting volume
4. Add the acid slowly while swirling until the indicator changes colour (end point)
 - Record the final volume
5. Repeat the titration without indicator, using the same measured volume of acid
6. Heat the neutral solution in an evaporating basin until concentrated
 - Check the solution is saturated by dipping a cold, glass rod into the solution and seeing if crystals form on the end
7. Leave the basin in a warm place to crystallize
8. If necessary, decant any excess liquid
9. Dry the crystals with filter paper



Your notes

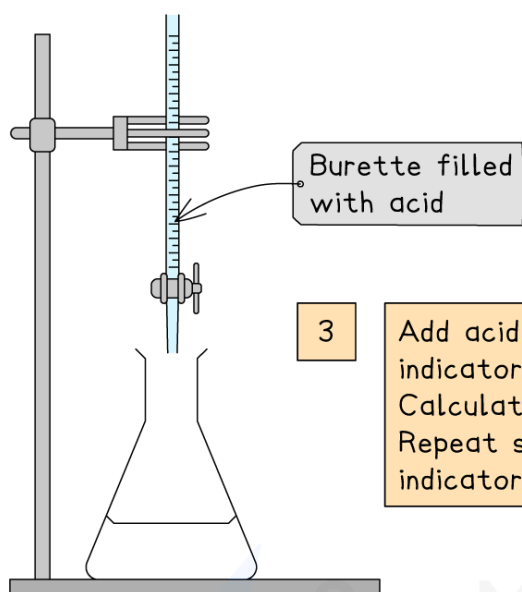
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Add alkali + indicator to conical flask using a pipette



2

Add acid to burette, noting the starting volume

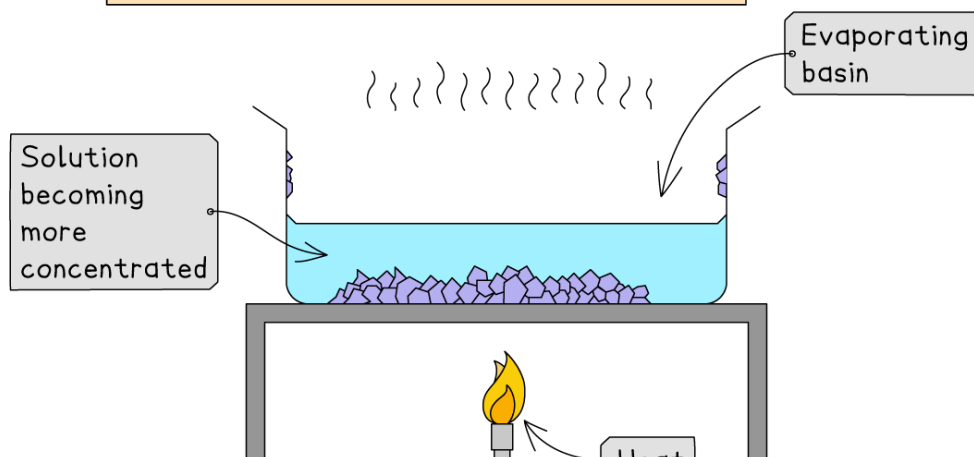


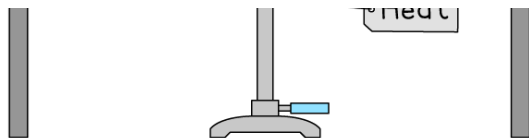
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Add acid to alkali slowly until indicator changes colour. Calculate volume of acid added. Repeat steps 1–3 without indicator.

4

Transfer solution to an evaporating basin, heat to partially evaporate water





Your notes

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Remove evaporating basin from heat and allow filtrate to dry and crystallize

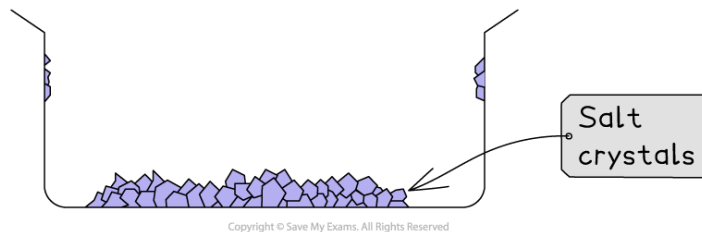
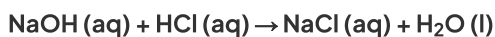


Diagram showing the apparatus needed to prepare a salt by titration

- Example reaction – preparing sodium chloride crystals:

sodium hydroxide + hydrochloric acid → sodium chloride + water



Examiner Tips and Tricks

Use **Method B (titration)** if the base is soluble (alkali)

Use **Method A (excess solid)** if adding a solid in excess, then filtering and crystallising:

- A1** for excess metal
- A2** for excess insoluble base
- A3** for excess insoluble carbonate



Preparing insoluble salts

Extended tier only

- Insoluble salts can be prepared using a **precipitation reaction**
- The solid salt obtained is the precipitate
- To successfully use this method:
 - The solid salt (precipitate) being formed must be insoluble in water
 - The reactants must be soluble

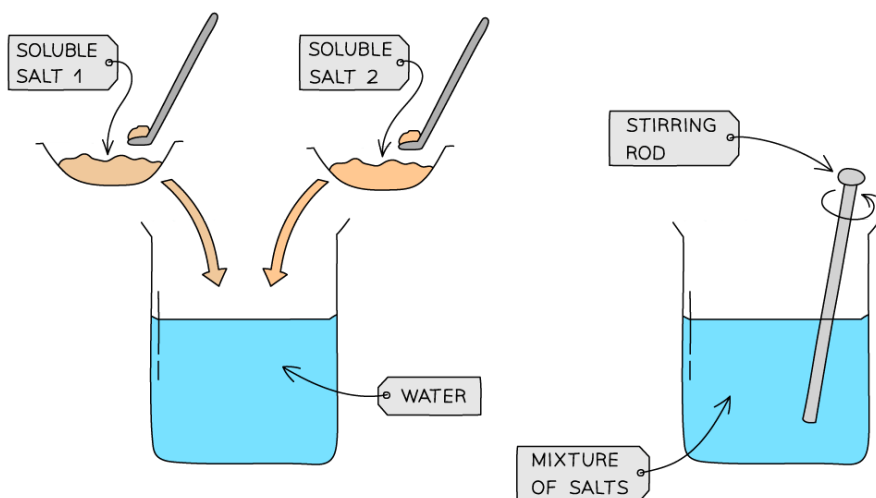
Using two soluble reactants



Your notes

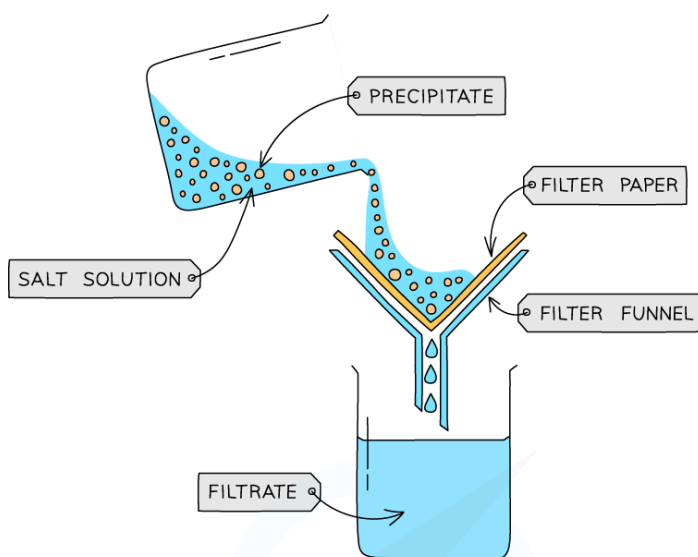
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ADD SOLUBLE SALTS TO WATER AND MIX



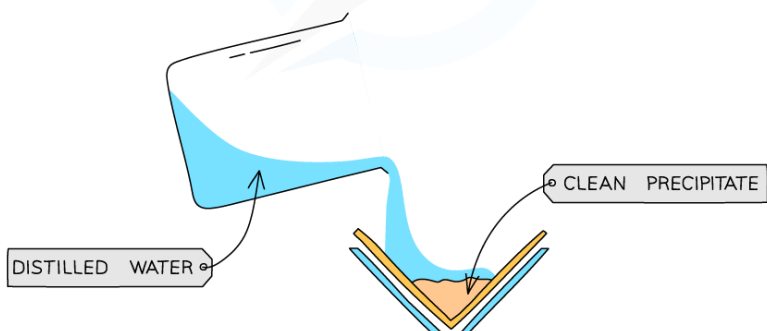
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FILTER TO REMOVE PRECIPITATE FROM THE MIXTURE



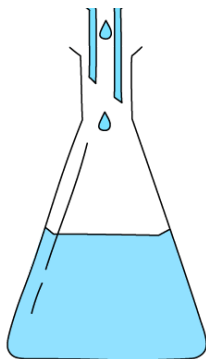
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WASH THE PRECIPITATE WITH DISTILLED WATER TO REMOVE TRACES OF SOLUTION



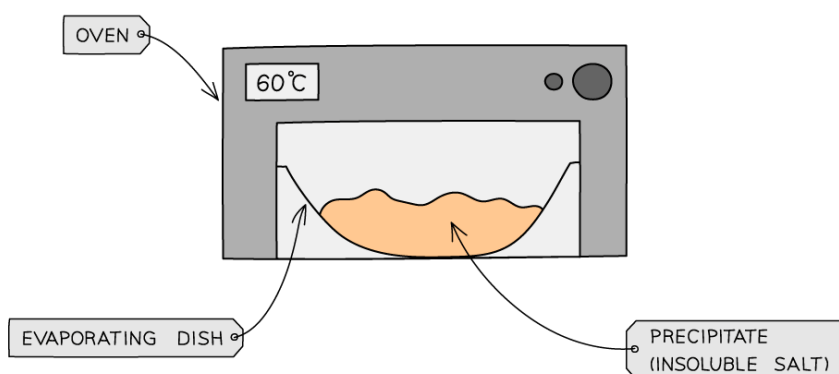


Your notes



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DRY THE PRECIPITATE (INSOLUBLE SALT) IN AN OVEN



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Diagram showing the filtration of the mixture to remove the precipitate

Method

- Make aqueous solutions of two soluble salts
- Mix together using a stirring rod in a beaker
- Filter to remove the precipitate from the mixture
- Wash the precipitate with distilled water to remove traces of other solutions
- Leave to dry (for example in a warm oven)

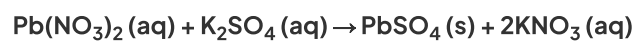
Example: Preparation of pure, dry, solid lead(II) sulfate using a precipitation reaction

- Dissolve lead(II) nitrate and potassium sulfate in water
- Mix together using a stirring rod in a beaker
- Filter to remove the precipitate from the mixture
- Wash the precipitate with distilled water to remove traces of potassium nitrate solution
- Leave to dry (for example in a warm oven)



Your notes

lead(II) nitrate + potassium sulfate → lead(II) sulfate + potassium nitrate





Solubility rules

- Salts are prepared by different methods, depending on whether the salt is soluble or insoluble so it is important to know the solubility of salts

Solubility of the common salts

Salts	Soluble	Insoluble
Sodium, potassium and ammonium	All	None
Nitrates	All	None
Chlorides	Most are soluble	Silver and lead(II)
Sulfates	Most are soluble	Barium, calcium and lead(II)
Carbonates	Carbonates of sodium, potassium and ammonium	Most are insoluble
Hydroxides	Hydroxides of sodium, potassium and ammonium (calcium hydroxide is sparingly soluble)	Most are insoluble



Hydrated & anhydrous salts

- When salts are being prepared, some water can be retained within the structure of the salt during the crystallisation process
- Salts that contain water within their structure are called **hydrated** salts
- Anhydrous** salts are those that contain no water in their structure
- A common example is copper(II) sulfate which crystallises forming the salt hydrated copper(II) sulfate, which is blue
- When it is heated, the water from its structure is removed, forming anhydrous copper(II) sulfate, which is white
- The hydrated salt has been **dehydrated** to form the anhydrous salt
- This reaction can be reversed by adding water to anhydrous copper(II) sulfate



Water of crystallisation

Extended tier only

- Water molecules included in the structure of some salts during the crystallisation process are known as **water of crystallisation**
- A compound that contains water of crystallisation is called a **hydrated compound**
- When writing the chemical formula of hydrated compounds, the water of crystallisation is separated from the main formula by a **dot**:
 - Hydrated copper(II) sulfate is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 - Hydrated cobalt(II) chloride is $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$
- The formula shows the number of moles of water contained within one mole of the hydrated salt:
 - Hydrated copper(II) sulfate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, contains 5 moles of water in 1 mole of hydrated salt
- A compound which doesn't contain water of crystallisation is called an **anhydrous compound**:
 - Anhydrous copper(II) sulfate is CuSO_4
 - Anhydrous cobalt(II) chloride is CoCl_2
- The conversion of anhydrous compounds to hydrated compounds is **reversible** by heating the hydrated salt:
 - Anhydrous to hydrated salt:

- $\text{CuSO}_4 + 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- Hydrated to anhydrous salt (by heating):
 - $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 + 5\text{H}_2\text{O}$



Your notes

Hydrated copper(II) sulfate and anhydrous copper(II) sulfate

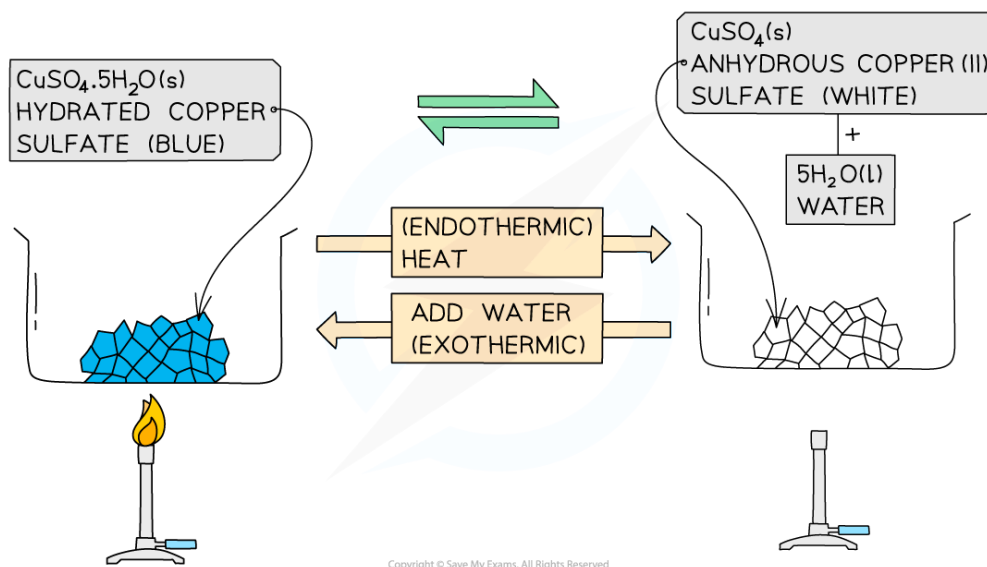


Diagram showing the dehydration of hydrated copper(II) sulfate